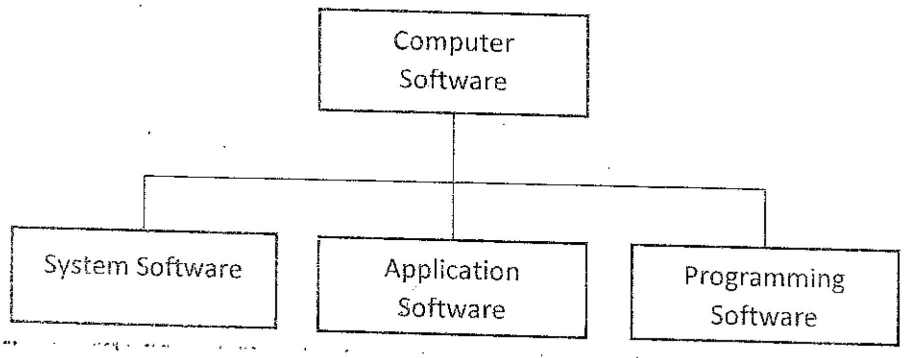
CHAPTER 3:

3.1 SOFTWARE SYSTEM

The software can be considered as a set of instructions, programs that are used to execute any particular task. The user cannot touch the software but can see through the GUI. The software can be considered as the variable part of the system while the hardware can be considered as an invariable part of the computer. And as there are many types of human language so is with the computer language also.

There are different types of computer languages present in the market. There are three types of software systems: system software, application, and programing sometime other includes driver's software.

Below are the types of Computer Software:



1. System Software

The system software is a type of computer software that is designed for running the computer hardware parts and the application programs. It is the platform provided to the computer system where other computer programs can execute.

The system software act as a middle layer between the user applications and hardware. The other purpose of system software is to translate inputs received from other sources and convert them into language so that the machine can understand.

- **The operating system** is the type of system software which is used to manage all other programs installed on the computer.
- The BIOS (basic input/output system) is another type of system software that works when the computer system starts and is used to manage the data between the hardware devices (video adapter, mouse, keyboard and printer) and the operating system. The system software provides the functionality for the user to use the hardware directly using the device drivers program.
- The boot is the system software program that loads the operating system in the main memory of the computer or can load in random access memory (RAM). The other example of system software is assembler which has a functionality to take computer instructions as input and then convert it into bits so that the processor can read that bit and perform computer operations.

The other example of system software is **a device driver** which is used to control some specific device which is connected to computer systems like mouse or keyboard. The device driver software is used to convert input/ output instructions of OS to messages so that the device can read and understand. The system software can be run in the background or can be executed directly by the user.

2. Application Software

The other category of software is application software that is designed for the users to perform some specific tasks like writing a letter, listening to music or seeing any video. For all these requirements there required a specific software for each type and that specific software that is designed for some specific purpose is known as application software. The operating software runs the application software in the computer system.

The difference between system software and application software

The difference between system software and application software is the difference in the user interface. In system software, there is no user interface present whereas in application software the user interface is present for each software so that users can easily use the software. The user cannot see the system software like an operating system and cannot work in system software but in an application software, users can see the application software using a graphical user interface and can also work in the application software. The user also has an option to create its user-written software and use the software for its personal use.

The templates are present which can be used by the user to create user-written programs. The application software can be bundled together and that bundle is known as an application suite. An example of an application suite is Microsoft Office application.

The Microsoft office software is designed by combining various small program to make one single program which can be used for writing text, creating a spreadsheet or creating presentations. The other type of application software is Mozilla Firefox, internet explorer. These kinds of application software can be used for searching any article, text on the web and interact with the outside world.

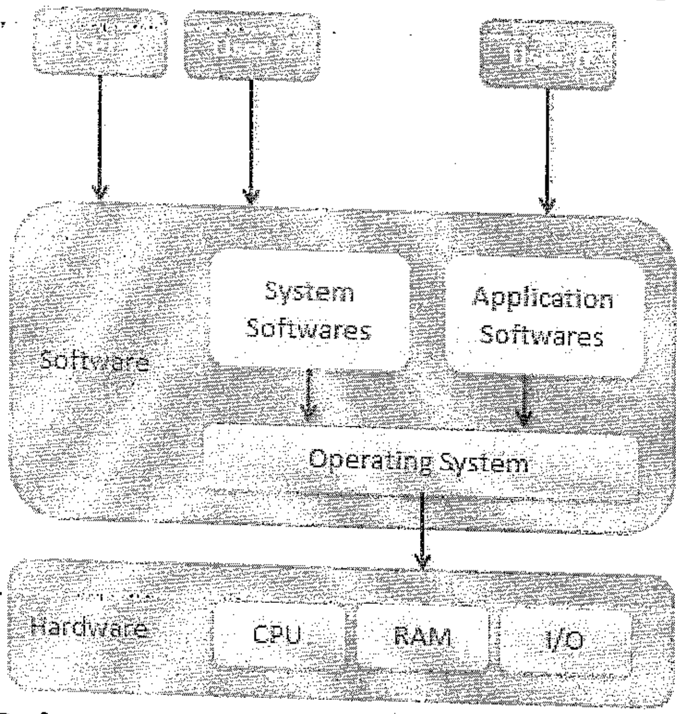
3. Programming Languages

The programming language is the third category of computer software which is used by the programmers to write their programs, scripts, and instructions which can be executed by a computer. The other name of the programming language is a computer language that can be used to create some common standards. The programming language can be considered as a brick which can be used to construct computer programs and operating system. The examples of programming languages are JAVA, C, C++, Python, Visual Basic and other languages.

There is always some similarity between the programming languages the only difference is the syntax of programming language which makes them different. The programmer uses the syntax and rules of programming language to write their programs.
Once the source code is written by a programmer in the IDE (Integrated Development. The use of programming language is in developing websites, applications, and many other programs.
The programming language can be broadly divided into two major elements syntax and semantics. The programming language follows some sequence of operations so that the desired output can be achieved. The programming language is also known as high-level language as the programs written by a programmer are easy to read and easy to understand. The JAVA, C, C++ programming languages are considered as high-level language. The other category of a programming language is a low-level language.

3.2 Operating System

An Operating System (OS) is an interface between a computer user and computer hardware. An operating system is a software which performs all the basic tasks like file management, memory management, process management, handling input and output, and controlling peripheral devices such as disk drives and printers. Some popular Operating Systems include Linux Operating System, Windows Operating System, VMS, OS/400, AIX, z/OS, etc.



Definition

An operating system is a program that acts as an interface between the user and the computer hardware and controls the execution of all kinds of programs.
Following are some of important functions of an operating System.

- Memory Management
- Processor Management

- Device Management
- File Management
- Security
- Control over system performance
- Job accounting
- Error detecting aids
- Coordination between other software and users.

A Multi-user operating system

Definition: A Multi-user operating system is a computer operating system which allows multiple users to access the single system with one operating system on it. It is generally used on large mainframe computers. Example: Linux, Unix, Windows 2000, Ubuntu, Mac OS etc., In the multi-user operating system, different users connected at different terminals and we can access, these users through the network ~~as shown in the diagram.~~

Features of the Multi-user Operating System

- Multi-tasking- Using multi-user operating system we can perform multiple tasks at a time, i.e. we can run more than one program at a time. Example: we can edit a word document while browsing the internet.
- Resource sharing- we can share different peripherals like printers, hard drives or we can share a file or data. For this, each user is given a small time slice of CPU time.
- Background processing- It means that when commands are not processed firstly, then they are executed in the background while other programs are interacting with the system in the real time.

Types of Multi-user Operating System

A multi-user operating system is of 3 types which are as follows:

1. Distributed Systems: in this, different computers are managed in such a way so that they can appear as a single computer. So, a sort of network is formed through which they can communicate with each other.
2. Time-sliced systems: in this, a short period is assigned to each task, i.e each user is given a time slice of the CPU time. As we know these time slices are tiny, so it appears to the users that they all are using the mainframe computer at the same time.
3. Multiprocessor Systems: in this, the operating system utilizes more than one processor. Example: Linux, Unix, Windows XP

Difference Between Single user and Multi user Operating System.

S/N	Single user Operating System	Multi-user Operating System
1	It is an operating system in which the user can manage one thing at a time effectively. Example: MS DOS	It is an operating system in which multiple users can manage multiple resources at a time Example: Linux,

		Unix, windows 2000, windows 2003 etc.
2	Single user Operating System has two types: Single user Single task Operating System and Single user Multi task Operating System.	It is of three types: time-sharing operating system, distributed operating system and multiprocessor system.
3	It is simple.	It is complex.
4	It provides a platform for one user at a time.	It provides controlled access for the number of users.
5	If another user wants to access the computer resources, then he/she has to wait until the current process completes.	There is no need to wait for accessing the computer resources.
6	In this, sometimes CPU is utilized to its maximum limit.	The operating system stimulates real-time performance by task switching.
7	It supports standalone systems.	It doesn't support standalone systems.
8	It is the operating system which maximum people use on their personal computers or laptops.	It is the operating system which is most of the time used in mainframe computers.
9	This type of operating system is used for single user.	This type of operating system is used for multiple users.
10	In this, there is no need to take care of balance between users.	In this, we have to take care of balance between users so that if one problem arises with one user does not affect other users also.

Advantages of the Multi-user Operating System

- When one computer in the network gets affected, then it does not affect another computer in the network. So, the system can be handled efficiently.

3.3 Job-shop

Job-shop scheduling, the job-shop problem (JSP) or job-shop scheduling problem (JSSP) is an optimization problem in computer science and operations research. It is a variant of optimal job scheduling. In a general job scheduling problem, we are given n jobs J1, J2, ..., Jn of varying processing times, which need to be scheduled on m machines with varying processing power, while trying to minimize the makespan – the total length of the schedule (that is, when all the jobs have finished processing). In the specific variant known as job-shop scheduling, each job consists of a set of operations O1, O2, ..., On which need to be processed in a specific order (known as precedence constraints). Each operation has a specific machine that it needs to be processed on and only one operation in a job can be processed at a given time. A common relaxation is the flexible job shop, where each operation can be processed on any machine of a given set (the machines in each set are identical).

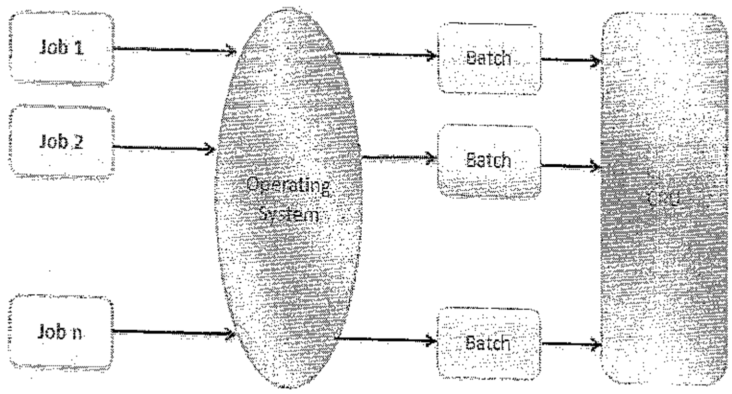
3.4 Batch processing

Batch processing is a technique in which an Operating System collects the programs and data together in a batch before processing starts.

Batch processing: The grouping together of several processing jobs to be executed one after another by a computer, without any user interaction. This is achieved by placing a list of the commands to start the required jobs into a BATCH FILE that can be executed as if it were a single program: hence batch processing is most often used in operating systems that have a COMMAND LINE user interface. Indeed, batch processing was the normal mode of working in the early days of mainframe computers, but modern personal computer applications typically require frequent user interaction, making them unsuitable for batch execution. Running a batch file is one example of batch processing, but there are plenty of others. When you select several documents from the same application and print them all in one step (if the application allows you to do that), you are "batch printing," which is a form of batch processing. Or let's say that you want to send a whole group of files to someone else via your modem-if your communications software permits batch processing, you can choose all the files you want to send, and have the software send them off in a batch while you go to the kitchen for a snack.

An operating system does the following activities related to batch processing –

- The OS defines a job which has predefined sequence of commands, programs and data as a single unit.
- The OS keeps a number a job in memory and executes them without any manual information.
- Jobs are processed in the order of submission, i.e., first come first served fashion.
- When a job completes its execution, its memory is released and the output for the job gets copied into an output spool for later printing or processing.



Advantages

- Batch processing takes much of the work of the operator to the computer.
- Increased performance as a new job get started as soon as the previous job is finished, without any manual intervention.

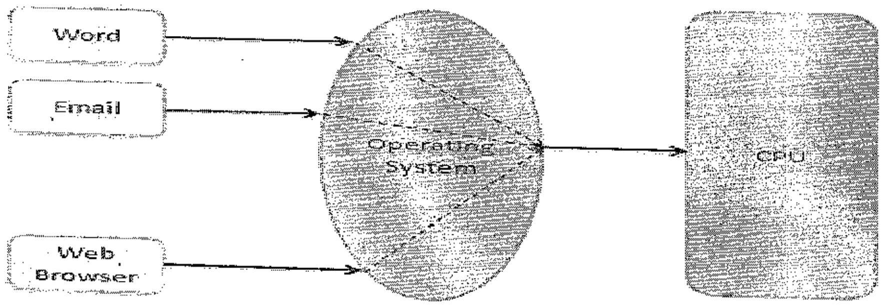
Disadvantages

- Difficult to debug program.
- A job could enter an infinite loop.
- Due to lack of protection scheme, one batch job can affect pending jobs.

3.5 Multitasking

Multitasking is when multiple jobs are executed by the CPU simultaneously by switching between them. Switches occur so frequently that the users may interact with each program while it is running. An OS does the following activities related to multitasking –

- The user gives instructions to the operating system or to a program directly, and receives an immediate response.
- The OS. handles multitasking in the way that it can handle multiple operations/executes multiple programs at a time.
- Multitasking Operating Systems are also known as Time-sharing systems.
- These Operating Systems were developed to provide interactive use of a computer system at a reasonable cost.
- A time-shared operating system uses the concept of CPU scheduling and multiprogramming to provide each user with a small portion of a time-shared CPU.
- Each user has at least one separate program in memory.
- A program that is loaded into memory and is executing is commonly referred to as a **process**.

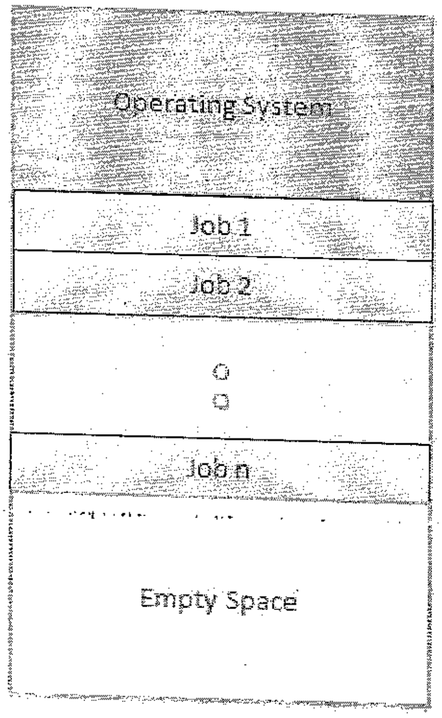


- When a process executes, it typically executes for only a very short time before it either finishes or needs to perform I/O.
- Since interactive I/O typically runs at slower speeds, it may take a long time to complete. During this time, a CPU can be utilized by another process.
- The operating system allows the users to share the computer simultaneously. Since each action or command in a time-shared system tends to be short, only a little CPU time is needed for each user.
- As the system switches CPU rapidly from one user/program to the next, each user is given the impression that he/she has his/her own CPU, whereas actually one CPU is being shared among many users.

3.6 Multiprogramming

Sharing the processor, when two or more programs reside in memory at the same time, is referred as **multiprogramming**. Multiprogramming assumes a single shared processor. Multiprogramming increases CPU utilization by organizing jobs so that the CPU always has one to execute.

The following figure shows the memory layout for a multiprogramming system.



An OS does the following activities related to multiprogramming.

- The operating system keeps several jobs in memory at a time.
- This set of jobs is a subset of the jobs kept in the job pool.
- The operating system picks and begins to execute one of the jobs in the memory.
- Multiprogramming operating systems monitor the state of all active programs and system resources using memory management programs to ensures that the CPU is never idle, unless there are no jobs to process

Advantages

- High and efficient CPU utilization.
- User feels that many programs are allocated CPU almost simultaneously.

Disadvantages

- CPU scheduling is required.
- To accommodate many jobs in memory, memory management is required.

3.7 Interactivity

Interactivity refers to the ability of users to interact with a computer system. An Operating system does the following activities related to interactivity –

- Provides the user an interface to interact with the system.
- Manages input devices to take inputs from the user. For example, keyboard. Manages
- output devices to show outputs to the user. For example, Monitor.

The response time of the OS needs to be short, since the user submits and waits for the result.

3.8 Computer Concepts – Programming Languages

A **program** is a set of instructions that help computer to perform tasks. This set of instructions is also called as scripts. Programs are executed by processor whereas scripts are interpreted. The languages that are used to write a program or set of instructions are called "**Programming languages**".

Programming languages are broadly categorized into three types –

- Machine level language
- Assembly level language
- High-level language

3.8.1 Machine Level Language

Machine language is lowest level of programming language. It handles binary data i.e. **0's** and **1's**. It directly interacts with system. Machine language is difficult for human beings to understand as it comprises combination of 0's and 1's. There is software which translate programs into machine level language. Examples include operating systems like Linux, UNIX, Windows, etc. In this language, there is no need of compilers and interpreters for conversion and hence the time consumption is less.

However, it is not portable and non-readable to humans.

```
0001111100001010101
0011100111001101010
0101010101010000000
1010101010101010101
1010100000111110000
1010101000111000101
1010101010010100100
```

Advantages

1. It makes quick and effective utilization of the machine

2. It obliges no interpreter to interpret the code i.e. directly understood by the machine

3.8.2 Assembly Level Language

Assembly language is a middle-level language. It consists of a set of instructions in a specific format called **commands**. It uses symbols to represent field of instructions. It is very close to machine level language. The computer should have assembler to translate assembly level program to machine level program. Examples include ADA, PASCAL, etc. It is in human readable format and takes lesser time to write a program and debug it. However, it is a machine dependent language.

Assembly Language	Machine Code
SUB AX, BX	0010101110000011
MOV CX, AX	100010111001000
MOV DX, 0	1011101000000000000000

Advantages Assembly Level Language

Below are the advantages:

1. It allows complex jobs to run in a simpler way.
2. It is memory efficient, as it requires less memory.
3. It is faster in speed, as its execution time is less.
4. It is mainly hardware-oriented.

3.8.3 High-level Language

High-level language uses format or language that is most familiar to users. The instructions in this language are called **codes** or **scripts**. The computer needs a compiler and interpreter to convert high-level language program to machine level language. Examples include C++, Python, Java, etc. It is easy to write a program using high level language and is less time-consuming. Debugging is also easy and is a human-readable language. Main disadvantages of this are that it takes lot of time for execution and occupies more space when compared to Assembly- or Machine-level languages. Following is a simple example for a high-level language –

```
if age < 18 {
  printf("You are not eligible to vote");
} else {
  printf("You are eligible to vote"); }
```

Advantages of a high-level language

- The high-level language is easy to read, write, and maintain as it is written in English like words.
- The high-level languages are designed to overcome the limitation of low-level language, i.e., portability.
- The high-level language is portable; i.e., these languages are machine-independent.

3.9 Compiler and Interpreter

3.9.1 Compiler

A compiler is a computer program that reads a program written in a high-level language such as FORTRAN, PL/I, COBOL, etc. It can translate it into the same program in a low-level language including machine language. The compiler also finds out the various errors encountered during the compilation of a program. The compiler converts high-level language into low-level language using various phases. A character stream inputted by the customer goes through multiple stages of compilation which at last will provide target language.

Advantages of Compiler

There are various advantages of the compiler which are as follows –

- A compiler translates a program in a single run. It consumes less time.
- CPU utilization is more.
- Both syntactic and semantic errors can be checked concurrently.
- It is easily supported by many high-level languages like C, C++, JAVA, etc.

3.9.2 Interpreter

An interpreter is a program that executes the programming code directly instead of just translating it into another format. It translates and executes programming language statements one by one. An interpreter takes less time to interpret a source program as distinguished by a compiler.

Advantages of Interpreter

There are various advantages of the interpreter which are as follows –

- An interpreter translates the program line by line.
- The interpreter is smaller in size.
- It is flexible.
- Error localization is easier.
- The interpreter facilitates the implementation of computer programming language constructs.